TITLE: Dense Multi-Vector Retrieval for Academic Literature Search
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ABSTRACT
We investigate dense multi-vector representations combined with late interaction cross-encoders for scientific research pap

1. INTRODUCTION
Scientific paper search requires precise keyword matching alongside semantic conceptual alignment. Traditional lexical app

2. METHODOLOGY & ARCHITECTURE
Our architecture utilizes a dual-encoder transformer pre-trained on contrastive citation graphs. For scoring, top candidates a

3. EXPERIMENTAL SETUP & DATASETS
We benchmark on the SciDocs and MS-MARCO academic subsets comprising 50,000 papers. Evaluation metrics include R

4. EXPERIMENTAL RESULTS

The cross-encoder reranker improves MRR@10 from 0.642 to 0.781 on SciDocs. BM25 alone achieves 0.589.

5. LIMITATIONS & FAILURE MODES

The authors explicitly acknowledge high computational latency during late-interaction cross-encoder inference (averaging 85

6. CONCLUSION & FUTURE WORK

Future work will explore sparse distillation and quantized vector embeddings to reduce memory requirements.